

Abstract—Abstract goes here.

Index Terms—COPD, wearable device, respiratory monitoring, human-robot interaction

I. INTRODUCTION

Chronic obstructive pulmonary disease (COPD) is a progressive chronic lung disease associated with persistent respiratory symptoms, such as breathlessness [1]. Activity related breathlessness is one of the most distressing and problematic symptoms experienced by people suffering from COPD, and is closely associated with reduced capacity for exercise and physical inactivity [2], [3] - an outcome that is of particular concern, as it has been shown in several studies that exercise can help alleviate the progression of COPD. Pulmonary rehabilitation (PR), incorporating exercise training and other patient-tailored interventions, can improve dyspnoea (breathlessness) related quality of life and exercise capacity in people with COPD [1].

Despite the impact of breathlessness on the quality of life of COPD patients, it is inherently subjective and thus difficult to quantifiably measure using physiological signals [2], [3]. Despite this, in recent studies, signals such as respiratory rate, heart rate and respiratory muscle activity have demonstrated associations with patient reported dyspnoea and may enable the estimation of breathlessness severity [4]. Wearable monitoring and telemonitoring technologies are also increasingly being studied within PR to collect physiological and symptom data, and to support personalised feedback [5].

II. LITERATURE REVIEW

Breathlessness is both subjective, and multidimensional in how patients experience it, and as such, severity is generally assessed through the self-reporting of patients, rather than through physiological measurements alone [2], [3]. Assessment of dyspnoea is done with a variety of methods: the mMRC scale describes limitation to function in daily life, while the Borg CR10 scale is often used to describe and quantify temporary breathlessness intensity during exercise [2]. Because the degree of breathlessness be altered by exertion, comparison of the intensity of symptoms is most useful when ratings are obtained at set points during a standardised exercise stimulus. Simultaneous physiological measurements can then be related to patient reported ratings [2], [3]. Despite the use of self-reporting, physiological signals still contain information that is associated with the degree of perceived dyspnoea. Dörssers et al. synchronised modified-Borg ratings with respiratory-muscle surface electromyography (EMG), respiratory rate (RR), heart rate (HR), oxygen saturation (SpO_2) and activity type during walking and cycling in 28 people with COPD [4]. A model containing RR, HR, activity type and SpO_2 gave fitted Borg

values within one point for 85.97 % of observations, increasing to 87.86 % after the inclusion of parasternal EMG. RR, HR and EMG were significant predictors to breathlessness intensity, while SpO_2 was found to not be independently significant. However, these findings are describing fitted associations, instead of externally validated and subject independent predictions. Additionally, high quality EMG was only available in 40 % of exercise segments. As such, the study supports the physiological estimation of breathlessness, but as of yet, not the reliable detection of breathlessness from a wearable device.

The feasibility of taking relevant measurements from the wrist has been investigated in other studies. Guber et al. compared wrist-based pulse oximetry with a more conventional fingertip reference in healthy participants with COPD [6]. Across 1,138 paired SpO_2 measurements, the wrist device achieved amplitude readings within 3 % of the reference values. However, measurements that were unreliable or collected during motion were excluded from analysis, and breathlessness was not assessed. As such, evidence that physiological signals can be related to breathlessness and evidence that those signals can be measured accurately at the wrist remain separate.

Existing rehabilitation technologies demonstrate that sensing can be integrated with personalised feedback and remote supervision. COPDTrainer used therapist-derived movement parameters and smartphone inertial sensing to give real time feedback on exercise performance during unsupervised COPD rehabilitation [7]. Carvalho et al. combined cardiorespiratory monitoring with breathing phase feedback and physiotherapist supervision in ten people with COPD, going to show the feasibility of integrating physiological sensing and task-specific guidance within pulmonary rehabilitation [8]. Otto-Yáñez et al. similarly demonstrated an eight-week supervised telerehabilitation programme incorporating repeated Borg assessment, wearable monitoring, symptom reporting and predefined recovery procedures/exercises [5]. However, only nine of the fifteen enrolled participants took the study to completion, and real time HR and SpO_2 monitoring during exercise was provided using primarily an O_2 ring, instead of a smartwatch.

Collectively, the literature supports patient reported assessment of exercise related breathlessness, the potential value of physiological and activity signals when correlated with dyspnoea, and the technical feasibility of wearable supported pulmonary rehabilitation. In combination with the clear problem of breathlessness experienced by COPD patients, the necessity for physician-prescribed exercises and the requirement to be able to complete these exercises at a moments notice - at home, in clinical settings, during exercise - creates a compelling argument to create and investigate new portable and effective methods of delivering these exercises and assisting users

through them.

III. CONCEPTUAL DESIGN

A. Concept Overview

The working concept is a wrist device with automatic prompting that guides a person with COPD through a breathing exercise during recovery after physical activity. Buttons allow the user to request assistance, respond to prompts and control the session, while haptic cues communicate the breathing routine. Wrist PPG and motion sensing will provide candidate physiological measurements and information about signal quality.

B. Design Requirements

The proposed device is intended to coach a person with COPD through a breathing exercise after physical activity or breathlessness. The suitable concept must meet the below requirements, with further detail and performance targets listed in Appendix A.

Under user interaction, the user must be able to start, pause and stop a session with minimal interaction, always. The device must also provide clear cues for the actions and timing of the breathing exercise. Finally, the user must be able to report breathlessness and other adverse symptoms easily.

For Monitoring & Reliability, the device should estimate and monitor the user's heart rate and respiratory rate when the available signals are sufficiently reliable. It must identify times where movement and other interference create unreliable data and avoid using that data for feedback and monitoring. Finally, the device should prompt an intervention if it detects breathlessness, or other concerning user states. See further details in Appendix A.

On top of these requirements, the device must also be comfortable to wear during exercise and long periods. It must also be battery powered, so users can go about their day. All the data must be stored locally on the device and require explicit consent from the user to share with clinicians.

C. Concept Selection and Rationale

The five concepts compared in Appendix B are a wrist-worn device providing manual timed guidance (C1), a wrist-worn device adding physiological monitoring (C2), a wrist-worn device adding automatic prompting (C3), a chest band with the same functionality of C2, and C3 with the addition of a pet robot frog as a physical coach.

We selected C3 because it combines monitoring with prompts to check symptoms and retains manual control. It has the added benefit over C1 and C2 of responding to the user without prompting, instead of needing user input for health benefits. This makes it a better health aid for users with COPD. We discounted C4 because of the reduced comfort and ease of user interaction daily, although a chest band would see more accurate measurements. The addition of a coach robot frog, which would mimic the breathing exercises for the user to follow along with would be novel, but has an unproven benefit to user health, and further hampers the day-to-day use of the device.

D. System Architecture and Operation

The system performs six functions: acquiring signals, assessing measurements, responds to assessment, reacts to user input, delivers breathing cues & stores relevant data locally. PPG provides biological sensing, while buttons and haptic feedback provide human-robot interaction through user input and device response.

Signal acquisition: The PPG sensor records an optical signal from the wrist for estimating heart rate and respiratory rate. The motion sensor records wrist movement to provide the user's activity for the context of the data. Appropriate filtering is applied to clean the signal, which is calibrated for each user's baseline during rest.

Measurement Assessment: The signals are then checked for interference, supplemented by the motion data. Unreliable measurements are discarded. The device then defines the user state based on the signal into resting, light activity, high activity, concerning heartrate or respiratory rate, and additionally breathlessness.

Assessment Response: The device uses these assessments to determine the feedback required. If the user records prolonged high activity, or breathlessness, the device will prompt the user to rest and begin a breathing exercise. If the user's heart rate or respiratory rate is concerningly high, based on the resting rate of the user, the device will prompt the user to rest and start a breathing exercise. Prompts request user confirmation before guidance starts, with the user able to dismiss them or request assistance independently.

UI Reaction: The device will also be constantly ready to respond to user interaction, allowing a user to start, stop and pause a breathing exercise, along with logging concerning symptoms like breathlessness, and score their symptoms regularly. The device will use this interaction to inform which exercise to guide the user through and for how long. The symptom scale is based on the Borg CR10 scale described in the literature review [2]: the user adjusts a 0–10 breathlessness rating with two buttons, receives vibration feedback for the selected value, and holds both buttons to confirm it.

Delivering Breathing Cues: The device stores several predefined breathing routines and will propose exercises which can be overruled by the user. It delivers haptic cues for the chosen routine's inhale, exhale and, where applicable, hold phases; pursed-lip breathing is the proposed default after activity. During still periods, wrist PPG may allow the device to estimate breathing phases and compare their timing with the cues, while motion sensing flags unreliable data. The user can pause or stop at any time.

Data Storage: The device stores selected data locally, like average heart rate and respiratory rate. It also stores symptom scores and logs, along with session details like duration and type of exercise. Unreliable data would not be stored. The data would only be able to be exported to a clinician or other medical professionals through a manual process requiring the user's explicit consent.

Operating sequence: The user can request assistance after activity or respond to an automatic prompt. Once resting, they

report their breathlessness and confirm whether to begin. The device delivers breathing cues and monitors usable signals while the user retains start, pause and stop control. If signals become unreliable, measurement-based feedback and automatic assessment are suspended; user-controlled timed guidance remains available. At completion or early stopping, the user is asked to report symptoms again and the session is stored locally.

IV. DETAILED DESIGN

This section presents the detailed mechatronic design of the proposed wearable device, building on the conceptual design outlined in Section III. The design is structured around several core aspects: the mechanical design of the wearable enclosure and its interface with the wrist, the electrical schematic and sensor circuitry required for physiological signal acquisition, the underlying software/control logic governing the device's operation, and the overall system architecture that integrates sensing, processing, and feedback into a cohesive device. Each of these aspects is discussed in turn, with the accompanying design rationale grounded in the requirements and constraints established in the preceding sections.

A. Mechanical Design

The wearable adopts a standard wristwatch form factor, comprising a single moulded housing and an adjustable strap to accommodate varying wrist sizes. As feedback is delivered exclusively through haptic actuation rather than a visual display, the design omits a watch face, simplifying the enclosure to a single sealed electronics compartment. TPU is used here, being a flexible, durable material commonly used in wearable enclosures. At the area of skin contact housing the PPG sensor, a clear polycarbonate window is integrated to permit optical transmission while protecting the underlying sensor from sweat and mechanical wear. Given the intended use during physical exercise, the device targets an IP67 ingress protection rating to ensure resistance to sweat and moisture. To achieve this while minimising assembly complexity, the housing is to be joined via ultrasonic welding, creating a continuous, seam-free seal without requiring additional sealing components such as O-rings or gaskets. The polycarbonate window is bonded to the housing using a UV-cured optical adhesive. Overall, the mechanical design follows established conventions for wrist-worn wearable devices, prioritising manufacturability and a proven form factor for user comfort. Fig. 1 shows an exploded view of the housing assembly, identifying the key mechanical and sealing components.

B. Electrical and Sensor Circuitry

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C. Software and Control Logic

hru

D. System Architecture

today

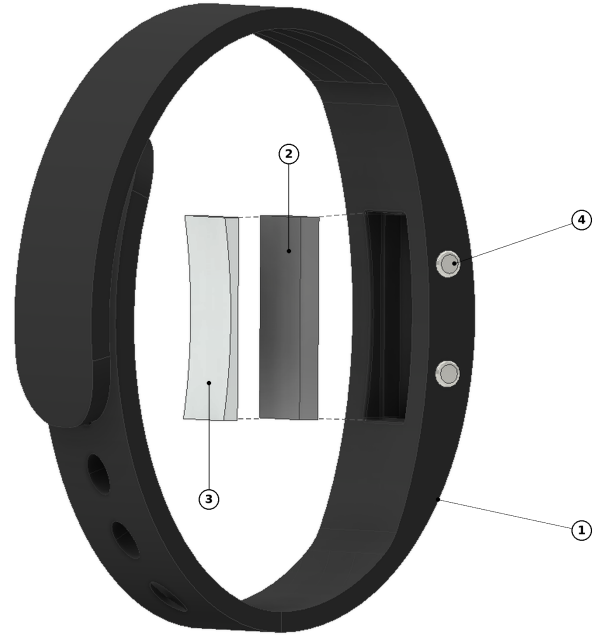


Fig. 1. Exploded view of the wearable housing assembly: (1) TPU housing with integrated watch strap, (2) electronics envelope (simplified), (3) polycarbonate optical window, (4) sealed push buttons.

V. VALIDATION OF FEASIBILITY

Much of the proposed device is already well supported by existing literature. The wristwatch form factor and photoplethysmography (PPG) sensing modality are widely adopted in consumer and research-grade wearables, and heart rate (HR) estimation from wrist PPG is a relatively mature problem, with numerous validated algorithms achieving high accuracy even during motion.

Respiration rate (RR) estimation from wrist PPG, however, remains an active area of research, largely because the respiratory-induced modulations of the PPG waveform are considerably weaker and more susceptible to corruption than the pulsatile component used for HR estimation. There is currently no consensus on the optimal signal processing pipeline for extracting RR from wrist PPG, and reliable RR estimation during motion, such as exercise, is essentially unsolved.

Given that the proposed device relies on RR as a key physiological input for estimating breathlessness, it is necessary to establish whether a reasonable RR estimate can be obtained from wrist PPG under realistic conditions before committing to this sensing approach. To this end, this section presents a validation analysis using raw PPG data from two randomly selected participants performing sitting and walking activities, drawn from the publicly available PPG-DaLiA dataset [9]. The two participants, referred to throughout by their dataset identifiers S7 and S8, also offer a useful contrast in self-reported baseline fitness: S7 is a 21-year-old female who reported rarely engaging in sport activity, while S8 is a 43-year-old male who reported exercising several times per week. This analysis does not attempt to solve the broader problem of reliable RR

estimation from wrist PPG during exercise, but instead aims to validate that a reasonable estimate can be obtained when compared against a ground truth respiration measurement. If reliable RR data can indeed be obtained in this way, then coupled with HR data, it would be recommended to conduct a further study into whether these two parameters alone form a sufficient basis to classify a patient's breathlessness severity, compared against their own self-assessment. Furthermore, despite this not addressing RR estimation from wrist PPG during motion itself, this is the intended use of the device: gathering this data during rest periods when the wearer is still, such as breaks during sports or rests between weightlifting exercises, to guide the wearer through breathing exercises that aid their recovery.

A. Ground Truth Derivation and the Effect of Motion

For each participant's sitting and walking sections, a respiration rate ground truth was derived from the chest-mounted RespiBAN unit's respiration (impedance pneumography) signal, rather than being taken from a single breath-detection algorithm. Three independent breath-detection methods – time-domain peak detection, zero-crossing counting, and spectral dominant-frequency estimation – were each run over 8 s windows (2 s shift) of the band-pass filtered (0.10–0.50 Hz) chest respiration signal, and a window's rate was only accepted once at least two of the three methods agreed to within 6 breaths/min; disagreeing or low-quality windows were rejected rather than reported. For the sitting sections, shown in Fig. 2, this ensemble produced an accepted rate for 97% (S7) and 89% (S8) of windows, with a smoothly varying respiration rate (mean 19.7 ± 1.4 and 18.9 ± 1.6 breaths/min respectively) consistent with expected resting adult respiration rates – giving confidence that this is a reasonable ground truth to validate the wrist PPG-derived estimate against.

The walking sections were considerably less reliable: the accepted-window rate dropped to 39% (S7) and 46% (S8), and the windows that were still accepted were far noisier (standard deviation rising from ~ 1.5 to ~ 5 breaths/min). Fig. 3 illustrates the source of this degradation at the wrist: the raw wrist PPG signal, which shows a clean, near-periodic pulse train of amplitude ~ 20 – 100 (a.u.) during sitting, is dominated by large, irregular motion artefacts (peaks exceeding ± 400) throughout the walking section, with no consistently identifiable pulse structure. Because the chest strap is more rigidly coupled to the torso than a wristband is to the wrist, its signal degrades less severely under motion, but even this reference becomes considerably less trustworthy while walking. Consistent with the literature discussed above, RR estimation from the wrist PPG during walking was therefore not pursued further in this validation; the remainder of this section focuses on the sitting data, where both the ground truth and the wrist-PPG-derived estimate are reasonably trustworthy.

B. Respiration Rate Estimation from Wrist PPG (Sitting)

For the sitting sections, respiration rate was estimated from the wrist PPG signal alone, with no reference to the chest strap,

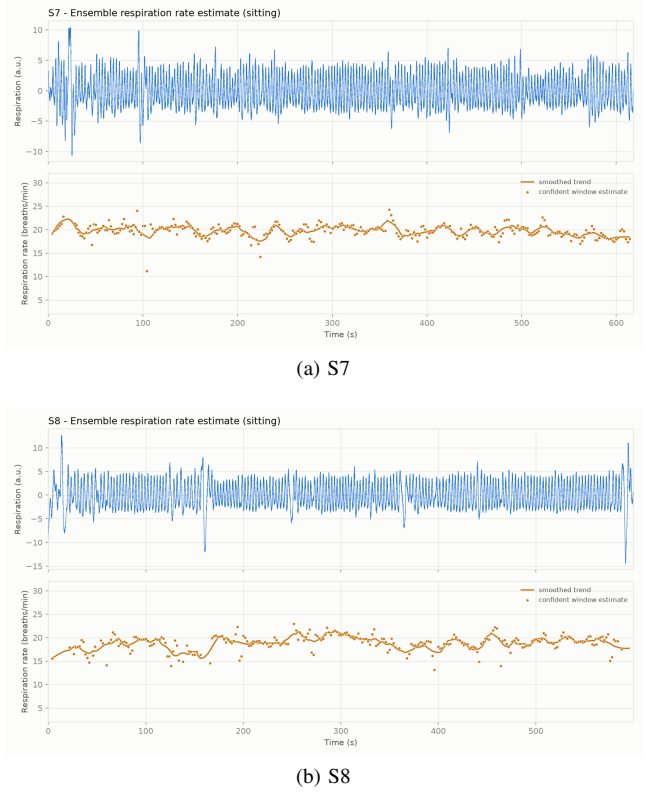


Fig. 2. Ground truth respiration rate estimation for the sitting sections of (a) S7 and (b) S8: raw chest RespiBAN respiration signal (top) and the resulting fused rate at each accepted window, with a smoothed trend (bottom). The high proportion of accepted windows (97% S7, 89% S8) and low estimate variance (± 1.4 – 1.6 breaths/min) support the use of this ensemble as ground truth for validating the wrist-PPG-derived estimate.

by exploiting three independent ways in which respiration modulates a PPG pulse train: frequency modulation of the pulse-to-pulse interval (respiratory sinus arrhythmia), amplitude modulation of pulse amplitude (respiration-driven changes in stroke volume), and baseline wander of the PPG's DC level (respiration-driven changes in peripheral blood volume). Individual heartbeats were first identified from the wrist PPG using an incremental-merge peak/onset detector, with beat-level rejection of implausible pulse intervals or amplitudes. Each of the three respiration-modulated streams was then reconstructed on a common time grid from these detected beats, band-pass filtered to the breathing band (0.10–0.50 Hz), and converted into a breathing rate every 2 s using peak-interval statistics from a rolling 24 s window; the three streams' rate estimates were fused with a quality-weighted vote, discarding a window's estimate where the streams disagreed excessively.

An initial version of this pipeline reproduced the correct within-window shape of the reference respiration trend but consistently underestimated its level, by around 3 breaths/min for S7 and 4 breaths/min for S8 – a fairly constant per-participant offset rather than one that varied over the session. This is expected: none of the three respiration-modulated streams carries any absolute information about a person's mean breathing rate, only about its relative fluctuation, so

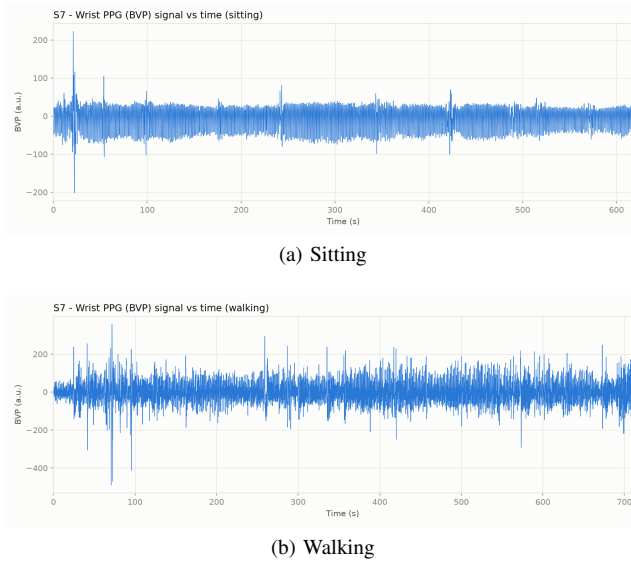


Fig. 3. Raw wrist PPG (BVP) signal for participant S7 during (a) sitting and (b) walking, over the same amplitude-comparable time windows. The near-periodic pulse train visible at rest is replaced by large, irregular motion artefacts once walking, with no consistently identifiable pulse structure – the reason wrist-PPG respiration estimation during motion was not attempted in this validation.

the fused estimate settles at whatever level its own beat-to-beat modulations happen to sit at. The pipeline was therefore revised to include a one-off per-participant DC baseline calibration: a single additive offset (+2.5 breaths/min for S7, +3.6 breaths/min for S8), fitted once against the reference respiration rate over the session, analogous to a short calibration a real device might perform once against a reference measurement before regular use. This step does not alter the underlying detection method or the breath-to-breath shape it recovers – only the constant level it is centred on.

Fig. 4 and Fig. 5, given in the appendix, show the resulting respiration rate for S7 and S8 respectively, compared against the RespiBAN ground truth over the full sitting section. After DC calibration, the mean absolute error against the ground truth fell from 3.1 to 1.8 breaths/min for S7, and from 4.3 to 2.2 breaths/min for S8 (RMSE: 3.8 to 2.3, and 5.0 to 2.8 breaths/min respectively). The estimate does not track every individual fluctuation in the reference trace – window-to-window correlation remains weak – reflecting how subtle the respiratory modulation of a wrist PPG pulse is relative to the noise floor discussed in the introduction to this section, but the overall level and slow trend are well matched throughout both six- to ten-minute sessions. As a secondary check, heart rate derived from the same wrist PPG beat detector was compared against the chest-strap reference heart rate over the same windows: for S7 the tracking was close (MAE 2.0 bpm), while for S8 it was periodically corrupted by isolated motion artefacts even during nominal sitting, producing brief overestimation spikes (up to 110 bpm against a true rate near 60 bpm) and a correspondingly higher MAE (9.6 bpm). Heart rate estimation is not the focus of this validation, but this nonetheless supports

the premise that pulse-level HR detection from wrist PPG is the more mature, lower-risk half of the sensing pipeline relative to RR.

C. Summary

This validation demonstrates that a reasonable respiration rate estimate can be derived from wrist PPG data alone, without any additional sensing modality, provided the wearer is reasonably still. It does not establish whether HR and RR alone are sufficient to predict patient-reported dyspnoea with strong accuracy; a further study, using a validated dyspnoea scale as ground truth, would be needed to determine this. It was also found that motion artefacts severely degrade the accuracy of the wrist-PPG RR estimate, a roadblock already well reported in the literature and consistent with the difficulty of RR estimation from wrist PPG during motion discussed at the outset of this section. The methodology used here is furthermore simple and illustrative rather than conclusive: a larger sample, spanning a range of genders, ages and activity levels across a wider range of activities, should be analysed before drawing firm conclusions, but this is outside the scope of the present report. The key takeaway is that signal processing of the raw wrist PPG signal is the most integral and challenging aspect of realising this product. This validation shows that it is achievable when the wearer is reasonably still – a condition well matched to the product’s intended use, in which the recovery phases of exercise (for example, breaks during sports or rests between weightlifting sets) are used to deliver guided breathing exercises tailored to the wearer’s own derived heart rate and respiration rate.

APPENDIX

Fig. 4 and Fig. 5 below are referenced from Section IV-B, and give the full per-participant wrist-PPG respiration rate results for S7 and S8 respectively.

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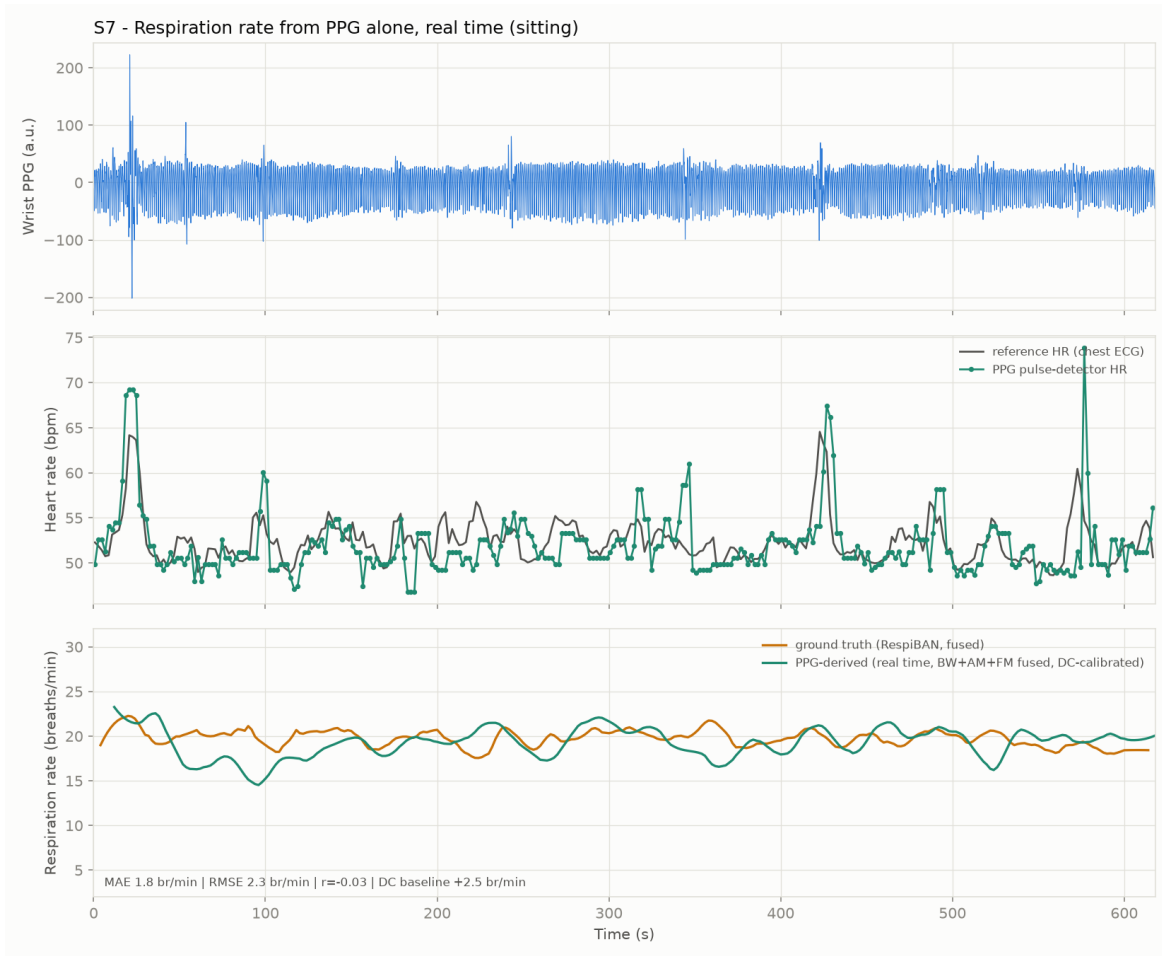


Fig. 4. Participant S7, sitting: wrist PPG (top); heart rate from the chest ECG reference vs. the wrist-PPG pulse detector (middle); and respiration rate from the RespiBAN ensemble ground truth vs. the wrist-PPG-derived estimate, after per-participant DC baseline calibration (bottom). The PPG-derived respiration rate closely tracks the reference level and slow trend (MAE 1.8 breaths/min, RMSE 2.3 breaths/min).

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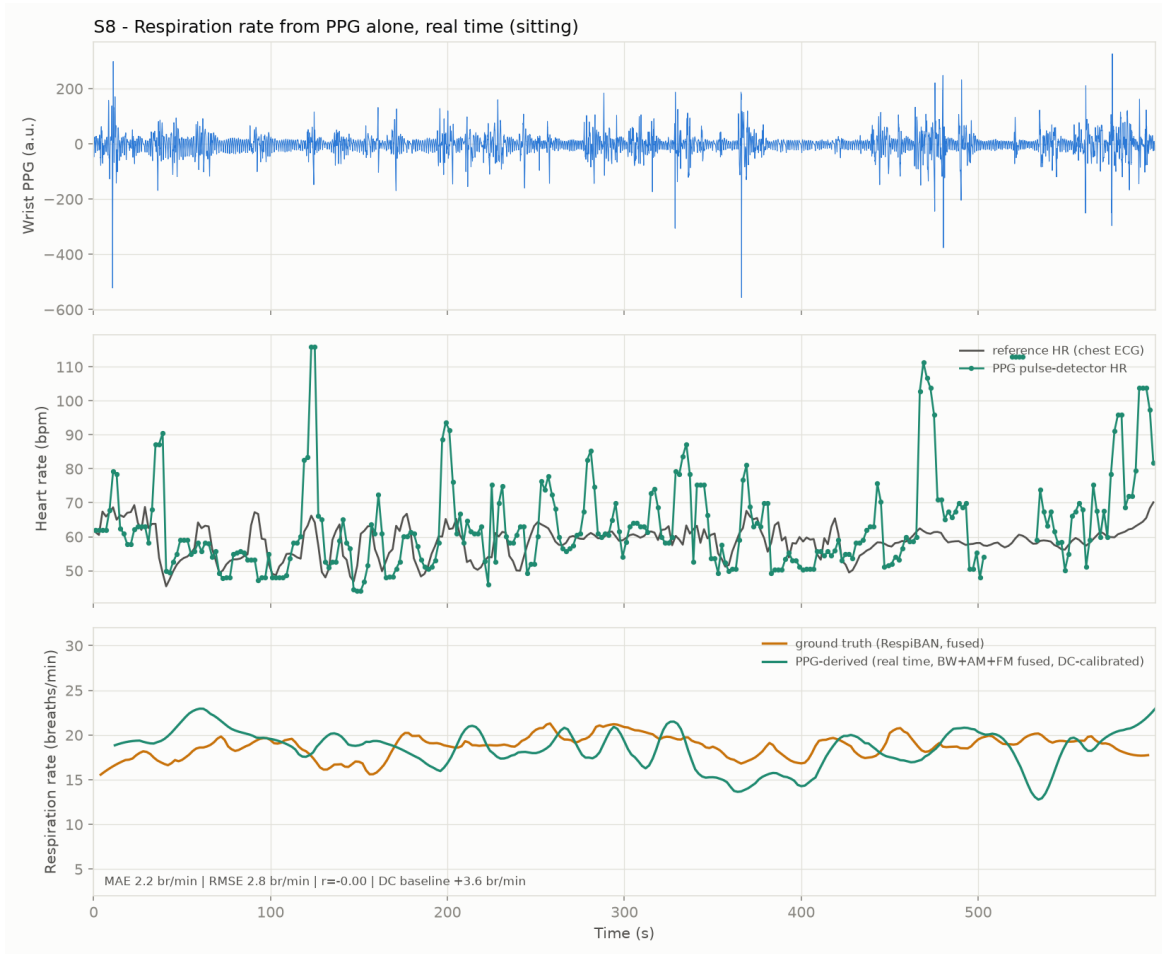


Fig. 5. Participant S8, sitting: wrist PPG (top); heart rate from the chest ECG reference vs. the wrist-PPG pulse detector (middle); and respiration rate from the RespiBAN ensemble ground truth vs. the wrist-PPG-derived estimate, after per-participant DC baseline calibration (bottom). The PPG-derived respiration rate again tracks the reference level and slow trend reasonably closely (MAE 2.2 breaths/min, RMSE 2.8 breaths/min), though heart rate tracking is more affected by intermittent motion artefacts than for S7.